Fiscal Year:	FY 2026	**Task Last Updated:**	FY 04/21/2026
PI Name:	Marquez, Jessica J. Ph.D.		
Project Title:	HCAAM VNSCOR: Crew Autonomy through Self-Scheduling: Guidelines for Crew Scheduling Performance Envelope and Mitigation Strategies		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jessica.j.marquez@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-6364
Organization Name:	NASA Ames Research Center		
PI Address 1:	Mail Stop 262-2, Building 262, Room 132		
PI Address 2:	Human Systems Integration Division		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	04/15/2019	**End Date:**	09/30/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
Contact Email:	alexandra.m.whitmire@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 per A. Beitman/JSC (Ed., 2/21/23)		
Key Personnel Changes/Previous PI:	February 2020 report: Mr. Steven Hillenius (Co-Investigator) left NASA. Dr. Tamsyn Edwards is replacing Mr. Hillenius as Co-I. Dr. Edwards works at NASA Ames as part of San Jose University Research Foundation. February 2021 report: Dr. John Karasinski is now a Co-I. February 2022 report: Dr. Edwards has left NASA. March 2023 report: Dr. Karasinski has changed affiliations. February 2024 report: Dr. Karasinski has changed affiliations.		
COI Name (Institution):	Bresina, John Ph.D. (NASA Ames Research Center) Gregory, Kevin M.S. (San Jose State University Research Foundation) Zheng, Jimin M.S. (San Jose State University Research Foundation) Edwards, Tamsyn Ph.D. (San Jose State University Research Foundation) Karasinski, John Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	This task is part of the Human Capabilities Assessments for Autonomous Missions (HCAAM) Virtual NASA Specialized Center of Research (VNSCOR). As NASA considers long-duration exploration missions (LDEMs), it is envisioned that crew will behave more autonomously as compared to low-Earth orbit missions. In this space environment, crew will have better and more timely insight as to how best to manage their own schedule, minimizing idle time as they wait for Mission Control Center (MCC) to respond or react to a delay in activity execution. Moreover, crew must be able to self-schedule, i.e., reschedule their own timeline without creating violations. NASA currently has not characterized crew performance for self-scheduling; specifically, non-expert human performance for the task of planning and scheduling has not been characterized experimentally. The focus of this proposal is to quantify crew performance envelope for the task of planning and scheduling as a function of plan complexity, and develop mitigations that are aimed at improving performance in the face of complex planning requirements. With regards to crew performance, we will study the relationship between planning efficiency, effectiveness, crew situation awareness, trust in planning software, and plan complexity. Once a performance envelope has been identified, we will shift our research emphasis to develop and evaluate countermeasures that mitigate adverse effects on performance. These mitigations will be evaluated in analogs and recommended countermeasures will be put forward if crew performance improves as compared to the baseline. Finally, based on research results, we will recommend corresponding standards and guidelines appropriate for autonomous crew in LDEMs.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	NASA currently has not characterized crew performance for self-scheduling; specifically, novice human performance for the task of planning and scheduling has not been characterized experimentally. As a result of this research, we will quantify the user performance envelope for the task of planning and scheduling, which impacts many jobs both on Earth and in spaceflight. The knowledge gained from our research can be generalized to benefit our understanding on how to improve roles that require planning and scheduling, such as project planning, personnel scheduling, and operational management. Our research will also contribute to developing the next generation of planning, scheduling, and execution software tools for NASA.
Task Progress:	As NASA considers long-duration exploration missions (LDEMs), it is envisioned that crew will behave more autonomously as compared to low-Earth orbit missions. In this space environment, crew will have better and more timely insight how best to manage their own schedule, minimizing idle time as they wait for Mission Control Center (MCC) to respond or react to a delay in activity execution. Moreover, crew must be able to self-schedule—that is, reschedule their own timeline without creating violations. NASA currently has not characterized crew performance for self-scheduling; specifically, non-expert human performance for the task of planning and scheduling has not been characterized experimentally. This research endeavored to contribute knowledge to this gap and evaluate the impact of self-scheduling in an analogous spaceflight environment. The focus of this research was to quantify crew performance as a function of scheduling task complexity. We studied the relationship between planning efficiency, effectiveness, workload, crew situation awareness, trust, and usability in scheduling software. To that effect, we conducted four lab experiments, examining the effect of number of activities (12, 24, 36), type of constraint (Time Range, Ordering, Claims, and Requires), percent of activities with constraints (0%, 33%, 66%), and task (scheduling vs. rescheduling). Based on controlled-lab studies, we found that for a fixed number of activities, the number of constraints significantly affects scheduling and rescheduling performance. However, a low number of constraints (1/3 of the activities having constraints) resulted in performance and workload that was no different than scheduling and rescheduling without constraints. We also found a main effect due to the type of constraint being scheduled—specific constraints affected performance differently. Time Range constraints took the longest while Claims constraints incurred the most violations while performing the task. When constraints involved two or more activities (Claims and Ordering), we detected lower situation awareness. Novel algorithmic analysis of scheduling task complexity was consistent with observed human performance. Overall, the software tool used for these experiments, Playbook, resulted in high trust and high usability scoring; evidence suggests a strong, significant correlation between learned trust and usability. Based on these results, we hypothesized that countermeasures would benefit performance associated with Time Range and Claims constraints. Several countermeasure aids were proposed and two were implemented and evaluated—No-Go Zones and Suggested Fixes. The fourth lab experimented aimed at specifically identifying the effect due to countermeasure presence for the scheduling task only. The presence of countermeasures improved scheduling performance, significantly lowering the number of violations created for all constraint types while also significantly lowering scheduling time for the Time Range constraint. The developed mitigation, i.e., countermeasure aids, improved performance in the face of complex scheduling requirements. Self-scheduling performance was also investigated and evaluated in an analogous spaceflight environment, the Human Exploration Research Analog (HERA). In HERA, we participated in Campaigns 6 and 7 (C6 and C7), which totaled eight missions. Our aim was to characterize self-scheduling and assess countermeasure aids that mitigate adverse effects on performance. In HERA C6 and C7, analog crewmembers were required to schedule one or two mission days for the entire team. Using the same scheduling software tool as in the control lab experiments, crew were similarly given a list of activities to schedule. Only half of the participants had access to the countermeasure. There was no significant effect detected due to presence of countermeasure on scheduling performance or workload. However, usability significantly increased for those that had the countermeasure. For C7, trust significantly increased over time and there was again a strong correlation between usability and trust at the end of the mission. Finally, a new survey to assess quality of self-scheduled plans, Plan Goodness, was validated in HERA C6 and C7. Content analysis on voice logs for HERA C6 and C7 were conducted. Two specific times were evaluated: team preference meeting discussions and allotted self-scheduling sessions. The first team preference meeting was the most valuable time for crew to actively discuss their schedules. Other allotted discussion times were not used as effectively. Surprisingly, we found that crew continued discussing timeline preferences during self-scheduling sessions despite this being an individual's task to complete. For C6, off-topic discussions were significantly correlated and inversely proportional with workload (though this effect was not seen in C7).

	HERA C6 and C7 concept of operations centered around crew autonomy. As such, a large portion of the scheduled HERA timeline had flexible activities, meaning, they could be rescheduled. Our experimental design did not require any of these additional activities to be self-scheduled, yet a large portion of them were. Spontaneous self-scheduling by the crew was present in all eight missions. The amount of self-scheduling was variable and not driven by the presence of countermeasure aids. Unfortunately, we have not been able to attribute the amount of self-scheduling performed to any individual or team factor. For C6, an exploratory analysis revealed that almost half the activities were flexible, and that from those, a fourth were rescheduled by crew. Three emergent behaviors were identified: rescheduling activities for the future and moved to "now;" rescheduling activities that were scheduled for "now" but moved to start later; and rescheduling activities that were for the future and had their start time slightly adjusted. Such characterization of crew autonomy at the mission operations level is unprecedented. Collectively, these behaviors suggest that crew self-scheduling is a desirable capability, has a low entry-barrier, does not add to crew's overall workload as it is regularly done, and moreover, is used to inform and accurately reflect current execution as well as projecting adjustments to schedule based on execution. With over 40 publications over seven years, this project's key outcomes center around human performance of novice schedulers and the impact of crew self-scheduling on concepts of operations centered on crew autonomy. Twenty-six standards and guidelines related to crew autonomy in LDEMS and self-scheduling are recommended for consideration by NASA.
Bibliography Type:	Description: (Last Updated: 05/06/2026)
Abstracts for Journals and Proceedings	Marquez JJ, Karasinski JA, Stirling L, Lee JD. "Space Human Factors Research: NASA's Human Capabilities Assessments for Autonomous Missions." 69th Human Factors and Ergonomics Society (HFES) International Annual Meeting, Chicago, IL, October 13-17, 2025. 69th Human Factors and Ergonomics Society (HFES) International Annual Meeting, Chicago, IL, October 13-17, 2025. , Oct-2025
Papers from Meeting Proceedings	Abbott R, Karasinski JA, Marquez JJ. "Characterizing Spontaneous Self-Scheduling in NASA's Human Exploration Research Analog Campaign 6." 46th IEEE Aerospace Conference, Big Sky, MT, March 1-8, 2025. 46th IEEE Aerospace Conference, Big Sky, MT, March 1-8, 2025. , Mar-2025
Papers from Meeting Proceedings	Vanderdonckt J, Vatavu RD, Manon J, Maddox R, Saint-Guillain M, Lefevre P, Marquez JJ. "UX, but on Mars: Exploring User Experience in Extreme Environments with Insights from a Mars Analog Mission." 2025 ACM Designing Interactive Systems Conference (DIS), Funchal, Madeira, PT, July 5-9, 2025. 2025 ACM Designing Interactive Systems Conference (DIS), Funchal, Madeira, PT, July 5-9, 2025. (pp. 3235-3250). , Jul-2025
Papers from Meeting Proceedings	Shelat S, Homer KE, Karasinski JA, Marquez JJ. "Multidimensional Usability Assessment in Spaceflight Analog Missions." SpaceCHI 2025, Cologne, Germany, June 23-24, 2025. SpaceCHI 2025, Cologne, Germany, June 23-24, 2025. , Jul-2025
Papers from Meeting Proceedings	Karasinski JA, Marquez JJ. "Analog Astronaut Self-Scheduling in NASA's HERA Campaign 7." 69th Human Factors and Ergonomics Society (HFES) International Annual Meeting, Chicago, IL, October 13-17, 2025. 69th Human Factors and Ergonomics Society (HFES) International Annual Meeting, Chicago, IL, October 13-17, 2025. , Oct-2025